b. Solve the following system of equations by Gauss-Seidal method correct to 5 decimal places $10x-5y-2z=3;\ 4x-10y+3z=-3;\ x+6y+10z=-3$.

29. a. The population of a town is as follows.

| Year | 1941 | 1951 | 1961 | 1971 | 1981 | 1991 |
|---|---|---|---|---|---|---|
| Population (in lakhs) | 20 | 24 | 29 | 36 | 46 | 51 |

Estimate the population during the period 1946 and 1976.

**(OR)**

b. Using Newton's divided difference formula, find the values of f(2), f(8) and f(15) from the following data.

| $x$ | 4 | 5 | 7 | 10 | 11 | 13 |
|---|---|---|---|---|---|---|
| f $(x)$ | 48 | 100 | 294 | 900 | 1210 | 2028 |

30. a. The population of a certain town is given below. Find the rate of growth of the population in 1931 and 1971.

| Year $(x)$ | 1931 | 1941 | 1951 | 1961 | 1971 |
|---|---|---|---|---|---|
| Population $(y)$ | 40.62 | 60.80 | 79.95 | 103.56 | 132.65 |

**(OR)**

b. By dividing the range into ten equal parts, evaluate $\int_0^{\pi} \sin x dx$ by Trapezoidal rule and Simpson's rule. Verify your answer with actual integration.

31. a.i. Using Taylor's series method, find the value $y(0.1)$ given $\frac{dy}{dx}=x^2+y^2$ and $y(0)=1$. (8 Marks)

ii. Given $y'=x^2-y,\ y(0)=1$ find the value of y(0.1) by improved Euler method. (4 Marks)

**(OR)**

b. Apply the fourth order Runge-Kutta method to find y(0.2) given that $y'=x+y,\ y(0)=1$ taking h = 0.1.

32. a. Solve by Crank-Nicholson method the equation $u_{xx}=u_t$ subject to $u(x,0)=0,\ u(0,t)=0$ *and* $u(1,t)=t$ for two time steps.

**(OR)**

b. Solve $\nabla^2 u=8x^2y^2$ for square mesh given u = 0 on the four boundaries dividing the square into 16 sub-squares of length 1 unit.

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**B.Tech. DEGREE EXAMINATION, MAY 2018**
First to Sixth Semester

15MA206 – NUMERICAL METHODS
*(For the candidates admitted during the academic year 2015 – 2016 onwards)*

**Note:**
(i) **Part - A** should be answered in OMR sheet within first 45 minutes and OMR sheet should be handed over to hall invigilator at the end of 45th minute.
(ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: Three Hours Max. Marks: 100

**PART – A (20 × 1 = 20 Marks)**
Answer **ALL** Questions

1. The condition for the convergence of Newton-Raphson method in solving $f(x)=0$ is
(A) $|f(x)f''(x)|<[f'(x)]^2$ (B) $|f(x)f'(x)|<[f''(x)]^2$
(C) $|f(x)f''(x)|>[f'(x)]^2$ (D) $|f'(x)f''(x)|<[f(x)]^2$

2. The order of convergence of Regula-Falsi method is
(A) –2.618 (B) 2.618
(C) 1.618 (D) –1.618

3. In Gauss-Elimination method, the coefficient matrix A of the system Ax=B is transformed into
(A) A upper triangular matrix (B) A lower triangular matrix
(C) A diagonal matrix (D) Unit matrix

4. The rate of convergence in Gauss-Seidal roughly ______ as fast as Gauss-Jacobi method.
(A) Thrice (B) Twice
(C) Four times (D) Five times

5. The parabola of the form $y=ax^2+bx+c$ passing through the points (0, 0), (1, 1) and (2, 20)
(A) $y=9x^2-8x$ (B) $y=9x^2+8x$
(C) $y=8x^2+9x$ (D) $y=8x^2-9x$

6. The nth divided difference of a polynomial is
(A) Zero (B) n
(C) Constant (D) n + 1

7. The operators Δ and E relates
(A) $\Delta=E-1$ (B) $\Delta=E+1$
(C) $E=1-\Delta$ (D) $E+\Delta=1$

8. The operator E is called
(A) Forward difference operator
(B) Central difference operator
(C) Backward difference operator
(D) Shifting operator

9. If $h=1$, $\Delta y_0 = 0.0244$, $\Delta^2 y_0 = -0.0003$, $\Delta^3 y_0 = 0$, then the value of $\left(\frac{dy}{dx}\right)_{x=x_0}$ is
(A) 0.02455
(B) 0.02275
(C) –0.0003
(D) 0.0232

10. Simpson's rule will give exact result if the entire curve $y=f(x)$ is itself a
(A) Ellipse
(B) Hyperbola
(C) Parabola
(D) Straight line

11. In Simpson's one-third rule, y(x) is a polynomial of degree _______.
(A) One
(B) Two
(C) Three
(D) Four

12. The error in the Trapezoidal rule is of the order
(A) $h^4$
(B) $h$
(C) $h^2$
(D) $h^3$

13. The method which coincides with the Talyor's series solution upto the term of $h^4$ is
(A) Fourth order Runge-Kutta method
(B) Milne's method
(C) Euler's method
(D) Improved Euler method

14. The improved Euler method is based on the averages of
(A) Points
(B) Slopes
(C) Tangents
(D) Normal

15. If $k_1 = -0.1$, $k_2 = -0.095$, $k_3 = -0.09525$, $k_4 = -0.090475$ and $y(0)=1$ in Runge-Kutta method of fourth order then y(0.1) is
(A) 0.09048375
(B) 0.09025146
(C) 0.823366
(D) 0.819025

16. To apply Milne's method, we need atleast _____ prior values of Y to get the required value of Y.
(A) 2
(B) 3
(C) 1
(D) 4

17. The Bender-Schmidt method is useful to solve _________ equation.
(A) Laplace equation
(B) Poisson's equation
(C) One dimensional heat equation
(D) Two dimensional heat equation

18. The Bender-Schmidt recurrence scheme is valid only if k =
(A) $\frac{a}{h^2}$
(B) $\frac{ah^2}{2}$
(C) $ah^2$
(D) $\frac{2h^2}{a}$



19. The Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ is of _________type.
(A) Elliptic
(B) Hyperbolic
(C) Parabolic
(D) circle

20. The partial differential equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x,y)$ is called
(A) Poisson's equation
(B) Laplace equation
(C) Heat equation
(D) Wave equation

**PART – B (5 x 4 = 20 Marks)**
Answer **ANY FIVE** Questions

21. Write the normal equations when the equation $y = ax^2 + bx + c$ is fit by the method of least squares.

22. State the conditions for the convergence of Gauss-Jacobi iterative method for solving a system of simultaneous algebraic equations.

23. Express $3x^3 - 2x^2 + 7x - 6$ in factorial polynomial taking h = 1.

24. Form the divided difference table for the following data.

| $x$ | 1 | 3 | 6 | 11 |
|---|---|---|---|---|
| $y$ | 4 | 32 | 224 | 1344 |

25. Evaluate $\int_{-3}^{3} x^4\, dx$ by Trapezoidal rule.

26. Using Euler's method solve numerically the equation $y' = x + y$, $y(0)=1$ at $x=0.2$. Taking h = 0.2

27. Classify the PDE $\frac{\partial^2 u}{\partial x^2} + 2\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0$.

**PART – C (5 x 12 = 60 Marks)**
Answer **ALL** Questions

28. a.i. Fit a second degree parabola to the following data.

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| $y$ | 2 | 6 | 7 | 8 | 10 | 11 | 11 | 10 | 9 |

(8 Marks)

ii. Find an iterative formula to find $\sqrt{N}$ (where N is a positive number). (4 Marks)

**(OR)**